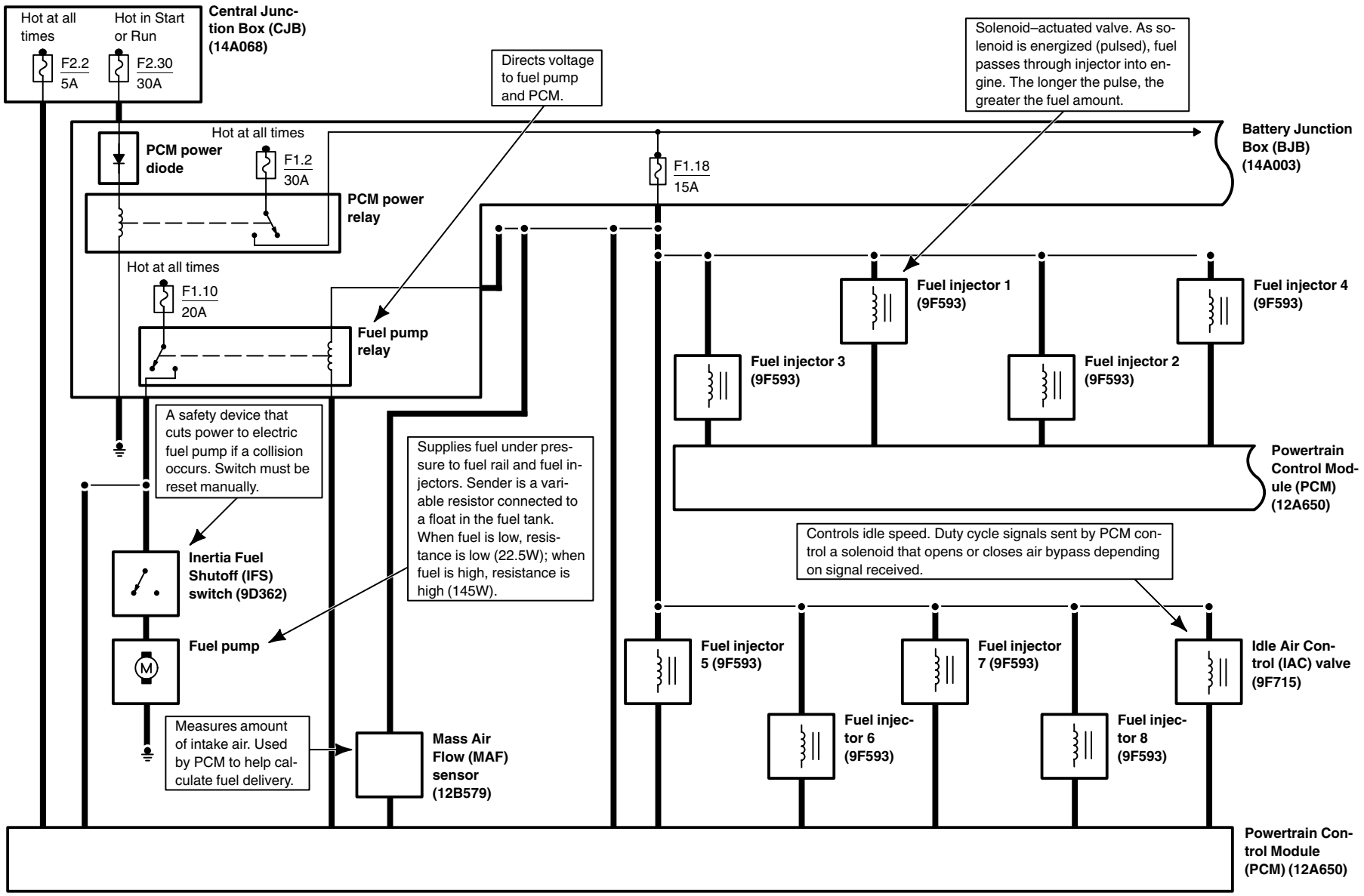5.4L, System overview